17 **(i)** Express $\dfrac{(x^2-8x+9)}{(x+1)(x-2)^2}$ in partial fractions. **[5]**

(ii) Express y in terms of x given that

$$\frac{dy}{dx} = \frac{y(x^2-8x+9)}{(x+1)(x-2)^2} \text{ and } y = 16 \text{ when } x = 3.$$

[7]